Collaborative Docs >

Discussion Agenda 08/11, 12/10/2018, 01/23/2019 (CGM+Clumps)

Below is a discussion agenda / live progress report for the Collaboration-wide brainstorming session of the 7th AGORA Workshop @ University of California Santa Cruz (Aug. 10-11, 2018; see the program, picture), and a telecon regarding Papers "CGM"/"Clumps" on Dec. 10, 2018.

7th AGORA Workshop participants:

• During this year's workshop, we will have the joint working sessions for Paper Groups “CGM” and “Clumps", with a focused, much-smaller-than-usual group of people — those who *actually* have carried out the 1e12 runs for the two Papers.

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• Everybody should feel free to edit the meeting notes, the content of which will eventually be merged into respective Paper Group Workspace.

Most recent changes that require your attention are in red color!

(1st revision: 08/15/2018, 2nd revision: 12/10/2018, 3rd revision 01/23/2019)

1. Paper Group Progress Report (as of Aug. 2018)

(1) Science-oriented Projects

<table>
<tr><th></th><th>Paper "CGM"</th><th>Paper "Clumps"</th><th>Paper "Metallicity"</th></tr>
<tr><td>GOALS</td><td>• Run ~1e12 Ms halo to study absorption lines</td><td>• Run ~1e12 (~1e13) Ms halo to study SF clumps at high z</td><td>• Run ~1e12 Ms halo to study metal distribution</td></tr>
<tr><td>LEADERS</td><td>• Cameron Hummels (Caltech)
• Santi Roca-Fabrega (UCM)</td><td>• Daniel Ceverino (Heidelberg)
• Alessandro Lupi (IAP)</td><td></td></tr>
<tr><td>PARTICIPANTS</td><td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Dekel, Mandelker, Primack, Roca-Fabrega, Strawn</td></tr>
<tr><td>ART-II</td><td>(Feldmann)</td></tr>
<tr><td>CHANGA</td><td>(Quinn)</td></tr>
<tr><td>ENZO</td><td>Hummels, Kim (Ji-hoon), Kim (Ki-won), O'Shea, Shin, Wise</td></tr>
<tr><td>GADGET</td><td>Nagamine, Shimizu</td></tr>
<tr><td>GASOLINE</td><td>Madau, Shen, Wadsley, Wissing</td></tr>
<tr><td>GEAR</td><td>Hausammann, Revaz</td></tr>
<tr><td>GIZMO</td><td>Dong, Hopkins, Hummels</td></tr>
<tr><td></td><td></td></tr>
</table></td><td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Ceverino, Dekel, Lee, Mandelker, Primack, Roca-Fabrega</td></tr>
<tr><td>ART-II</td><td>(Feldmann)</td></tr>
<tr><td>CHANGA</td><td>(Quinn)</td></tr>
<tr><td>ENZO</td><td>Kim, O'Shea</td></tr>
<tr><td>GADGET</td><td>Nagamine, Shimizu</td></tr>
<tr><td>GASOLINE</td><td>Mayer, Wadsley</td></tr>
<tr><td>GEAR</td><td>Hausammann, Revaz</td></tr>
<tr><td>GIZMO</td><td>Lupi, Hopkins</td></tr>
<tr><td>RAMSES</td><td>Agertz, Roca-Fabrega</td></tr>
</table>
(Code leaders as of Aug. 2018)
- ICs help: Buehlmann, Hahn
- Analysis help: Chakrabarti, Kim, Turk</td><td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>GADGET</td><td>Chakrabarti, Lewis</td></tr>
<tr><td>GEAR</td><td>(Revaz)</td></tr>
<tr><td>RAMSES</td><td>(Pallottini, Teyssier)</td></tr>
</table>
- Analysis help: Kim

• Working Groups engaged:
- IV (common analysis)
- IX (char. of cos. galaxies)
- XII (interstellar medium)</td></tr>
</table>

	RAMSES | Agertz, Roca-Fabrega, Teyssier (Code leaders as of Jan. 2019) - ICs help: Buehlmann, Hahn, Simpson - Analysis help: Chakrabarti, Fumagalli, Hummels, Kim, Madau, Roca-Fabrega, Smith, Turk • Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - V (subgrid physics) - IX (char. of cos. galaxies) - X (outflows)	• Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - XI (high-z galaxies)	
PROGRESS MADE OR BEING MADE	• Visit **Paper's Workspace** for the latest updates, and the Overleaf draft being written. • **Task Force** participants have completed or are about to complete the action items to jointly launch Papers "CGM" and "Clumps". We have strategically decided to finish Paper "CGM" first, then Paper "Clumps". See the Papers' Workspace for timeline details. • Code leaders have submitted preliminary datasets for the proposed cosmological run aimed at calibrating the feedback at z=4 first. Composite figures in the **Paper "CGM" Workspace**. • Code leaders are asked to write about their "favorite" feedback/cooling of their dataset entries in Section "3.2. Code-dependent Physics" of the draft. Overleaf link in the **Paper "CGM" Workspace**. • Hummels has updated the **ipython notebook** that explains how to use yt+Trident on datasets from various codes. Trident is now fully functional with "demeshened" version of yt (i.e., **yt-4.0** branch) as of Jun. 2018. More information in the **Paper "CGM" Workspace**.	• Visit **Paper's Workspace** for the latest updates, and the Overleaf draft being written. • **Task Force** participants have completed or are about to complete the action items to jointly launch Papers "CGM" and "Clumps".	• Visit **Paper's Workspace** for the latest updates. • Will piggyback on the planned Paper "CGM" runs and evolve them further down to z=0.
	• **Paper Workspace**	• **Paper Workspace**	• **Paper Workspace**

OTHER LINKS	• Trident download and documentation • Trident method paper (Hummels et al. 2016)		

Cf. other science-oriented project ideas: see this link

(2) Previous Projects

<table>
<tr><th></th><th>Paper "DM-Highres"</th><th>Paper "NoSubgrid"</th><th>Paper "Disk-II"</th></tr>
<tr><td>GOALS</td><td>• Run Proof-of-concept-DM (~1e11 Ms halo) at higher res</td><td>• Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)</td><td>• Compare simulation codes using common disk ICs</td></tr>
<tr><td>LEADERS</td><td>• Oliver Hahn (OCA, Nice)
• Christine Simpson (HITS)</td><td>• Nick Gnedin (Fermilab)
• Piero Madau (UCSC)
• Britton Smith (SDSC)</td><td>• Santi Roca-Fabrega (UCM)</td></tr>
<tr><td>PARTICIPANTS</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Ceverino, Klypin, Primack</td></tr>
<tr><td>ART-II</td><td>Gnedin, Feldmann, Li, Semenov (Leitner)</td></tr>
<tr><td>CHANGA</td><td>Quinn, Wallace</td></tr>
<tr><td>ENZO</td><td>Hummels, Kim, Simpson, Wise</td></tr>
<tr><td>GADGET</td><td>Choi, Nagamine, Onorbe, (Rocha)</td></tr>
<tr><td>GADGET-AFS/CFS</td><td>(Hobbs)</td></tr>
<tr><td>GASOLINE</td><td>Madau, Shen, (Pillepich, Zolotov)</td></tr>
<tr><td>GCD+</td><td>Kawata</td></tr>
<tr><td>GIZMO-AFS</td><td>Garrison-Kimmel, Lupi, Hopkins, Onorbe</td></tr>
<tr><td>PKDGRAV2</td><td>(Guedes, Kuhlen)</td></tr>
<tr><td>RAMSES</td><td>Hahn, Roca-Fabrega, Teyssier</td></tr>
</table>
(Code leaders as of Aug. 2016)
- Others interested: Dekel, Feldmann, Kawata, Pallottini, Robles</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Ceverino, Klypin</td></tr>
<tr><td>ART-II</td><td>Gnedin, Feldmann, Leitner</td></tr>
<tr><td>CHANGA</td><td>Quinn</td></tr>
<tr><td>ENZO</td><td>Hummels, Kim, Simpson, Smith, Turk, Wise</td></tr>
<tr><td>GADGET</td><td>Choi, Nagamine</td></tr>
<tr><td>GADGET-SPHS</td><td>(Hobbs)</td></tr>
<tr><td>GASOLINE</td><td>Keller, Madau, Mayer, Shen, Wadsley</td></tr>
<tr><td>GEAR</td><td>Revaz</td></tr>
<tr><td>GIZMO</td><td>Lupi, Thompson</td></tr>
<tr><td>RAMSES</td><td>Agertz, Butler, Teyssier</td></tr>
</table>
(Grackle implementers acknowledged in Paper "4")</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Arraki, Ceverino, Klypin, Mandelker, Snyder</td></tr>
<tr><td>ART-II</td><td>Feldmann, Gnedin, Leitner, Li, Semenov</td></tr>
<tr><td>CHANGA</td><td>Quinn, Wallace</td></tr>
<tr><td>ENZO</td><td>Abel, Butler, Butsky, Forbes, Hummels, Kim, O'Shea, Simpson, Turk, Wise</td></tr>
<tr><td>GADGET</td><td>Choi, Nagamine, Chakrabarti, Shimizu, (Todoroki)</td></tr>
<tr><td>GADGET-SPHS</td><td>(Hobbs)</td></tr>
<tr><td>GASOLINE</td><td>Benincasa, Keller, Madau, Mayer, Shen, Wadsley</td></tr>
<tr><td>GEAR</td><td>Revaz</td></tr>
<tr><td>GIZMO</td><td>Lupi, Hopkins, Thompson</td></tr>
<tr><td>RAMSES</td><td>Agertz, Butler, DeGraf, Teyssier, Nickerson</td></tr>
</table>
(Code leaders as of Aug. 2016 for Paper "4" before)</td></tr>
<tr><td></td><td>• Visit Paper's Workspace where Simpson, Hahn and others have been updating the progress.</td><td>• Code groups such as ART-I and GIZMO have carried out cosmological simulations with various AGORA ICs, even including sophisticated subgrid physics.</td><td>• Visit Paper's Workspace where the results for disk comparison are compiled (including those presented in our first disk comparison paper - previously</td></tr>
</table>

PROGRESS MADE OR BEING MADE	• We will plan to have 5-code comparison. As of Aug. 2017, 4+ participating groups have completed and submitted the production-level results (~100 pc resolution) to NERSC. • We may focus on the (survival of) substructures. This subset of topics could potentially to be led by Jiang as a smaller paper. • Consistent tree - python interface has been built by Britton Smith (ytree).		known as **Paper "4"**). The **agora-analysis-script** repository has been accordingly updated. • Analysis items that were not covered in our first disk comparison paper (see **this link**) will be considered in the next paper, Paper "Disk-II". • A subset of topics could be discussed in the "Metal Diffusion" or "Secular evolution" paper (see **this link**).
OTHER LINKS	• Progress report in **Paper's Workspace** and/or in **WG III's Workspace**. • **MUSIC Google group** • **ytree documentation**	• **GRACKLE Google group** • **GRACKLE download & documentation** • **GRACKLE method paper (Smith et al. 2017)** • **AGORA API for GRACKLE by Nick Gnedin**	• Progress report in **Paper's Workspace** and/or in **WG V's Workspace**. • **yt-AGORA Google group**

2. News and Updates

(1) AGORA Workshop 2019

• Next year's annual Santa Cruz Galaxy Workshop will be at UC Santa Cruz, August 5-9, 2019. The concluding day (Friday) of the Galaxy Workshop will be the first day of the 8th AGORA Workshop, August 9-10, 2019. We thank the continuous logistical support by Joel Primack and UC Santa Cruz.

(2) AGORA NERSC Resources

• We thank the annual effort by Joel Primack and Tom Abel to secure a continuous access to the NERSC computing resources for AGORA. The latest allocation for AY19 (Jan. 8, 2019 - Jan. 13, 2020) is 15 TB per project/archive storage each with half a million computing hours.

3. Recaps

(1) AGORA SMBH Prescription for Accretion and Feedback:

- Please visit **this link**.

(2) Strategy to Store Simulation Outputs:

- Please visit **this link**.

(3) How to request an account on NERSC:

- If you represent a code group but don't have a NERSC user account, please visit **this page**. Select "standard" account and find the repository name "agora". Joel Primack will add your name to the group when approved.

(4) Your Own yt Installation on NERSC:

- Please visit **this link**.

(5) yt-Sunrise Pipeline for AGORA on NERSC / Working Installation of Sunrise as Docker Image:

- Please visit **this link**.

(6) Mini-workshop Opportunities:

- Mini-workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the **yt-AGORA mini-workshop** at UCSC in Mar. 2014.

(7) Other Useful Links:

- Quick links to the important pages in AGORA Workspace, including internal policies: **here**
- Internal policy regarding the disclosure of AGORA simulation results: **here**
- Collection of softwares relevant to AGORA effort: **here**

4. Your Feedback

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the **mailing list**. You can also directly contact the **project coordinator** anytime!